

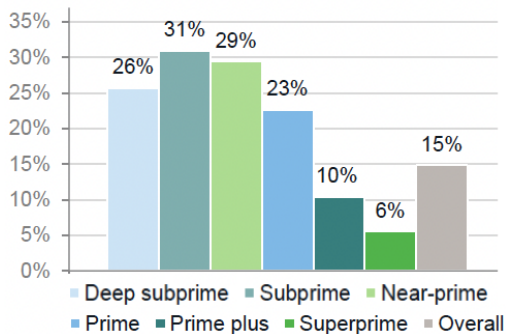
Targeting Higher Credit Card Payments

Benedict Guttman-Kenney

Rice University

Rice Finance Brownbag, 6th February 2026

Credit Cardholders Often Pay At Or Near Minimum Payment. Why?



(Consumer Financial Protection Bureau, 2025)

- 29% of accounts pay at or near the minimum payment

(Keys & Wang, 2019 JFE)

One Potential Explanation: Anchoring Heuristic

Anchoring & adjustment heuristic: Foundational bias in Tversky & Kahneman (1974)

Key idea:

- Anchors: Salient numbers that can bias consumer choices.
- Consumers choose at / near numerical amount initially presented, & adjust away too little.

“One of the most robust phenomena studied across the social behavioral sciences.”

- 50 year meta-analysis (Schley and Weingarten, 2025)

Anchoring Influential In The Credit Card Literature

Research across the fields of economics, finance, management, marketing, and psychology documents **“credit card anchoring”**

- Credit card minimum payment “can serve as an anchor, and as a nudge that this minimum payment is an appropriate amount” (Sunstein & Thaler, 2008)

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- Support for anchoring from studies of credit card administrative data (Keys & Wang, 2019; Medina, & Negrin, 2022; Vihriälä 2025)
 - Guttman-Kenney et al. (2025) shows anchoring does not explain autopay choices

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Key implication: Without anchoring, cardholders would ↑ credit card payments

This Paper: Do Online Experimental Findings Extrapolate To The Field?

Field experiment on 6,714 credit cardholders with a U.S. FinTech lender

- RCT varying in-app credit card payment options
 - Test shrouding minimum payment amount
 - Test adding 50% payment amount option
- Observe outcomes for six cycles

This Paper: Do Online Experimental Findings Extrapolate To The Field?

Field experiment on 6,714 credit cardholders with a U.S. FinTech lender

- RCT varying in-app credit card payment options
 - Test shrouding minimum payment amount
 - Test adding 50% payment amount option
- Observe outcomes for six cycles

Key Findings: Results of hypothetical online experiments do not translate into the field

- No increase in payments from shrouding the minimum payment
- Adding 50% payment option does not increase payments

Contributions

- Evidence that a **policy** shrouding the minimum payment is expected to be ineffective
 - (Financial Conduct Authority, 2021; Brookings, 2022; Financial Health Network, 2023)

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- **Household Finance:** Why do low score consumers often pay at/just above the minimum?
 - *Not* driven by heuristic anchoring to the minimum payment
 - Cardholders frequently lack liquid cash (but not at borrowing constraint)

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 - Target paying at least the minimum payment, which acts as a reference point
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- **Behavioral Economics/Finance:** Weakness of active choice nudges & anchoring heuristic

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 - Bunching makes sense given discontinuity in costs at minimum
- **Behavioral Economics/Finance:** Weakness of active choice nudges & anchoring heuristic
- **Empirical methods** in social sciences & challenges of scaling (List, 2022)
 - Online studies increasingly used across social sciences (Rauf et al., 2025)
 - Magnitudes can differ between online and field experiments (Gandhi et al., 2025)
 - Example where not just magnitudes differ, but key finding

1. The Experiment (Design, Data, Methodology)
2. Experimental Results
3. Heterogeneity, Distinguishing Anchoring from Targeting

1. The Experiment

Field Experiment Design

Control (C)

Make Payment - Current Balance (Default)

9:41

< Choose Amount

👉 Paying your current balance avoids interest, boosts your credit score, and lowers utilization.

\$124.00

Your statement balance is due on April 24

\$124.00 👉 Current Balance	\$74.00 Statement Balance
\$37.00 50% of Statement	\$25.00 Minimum Amount

Next

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Treatment T2

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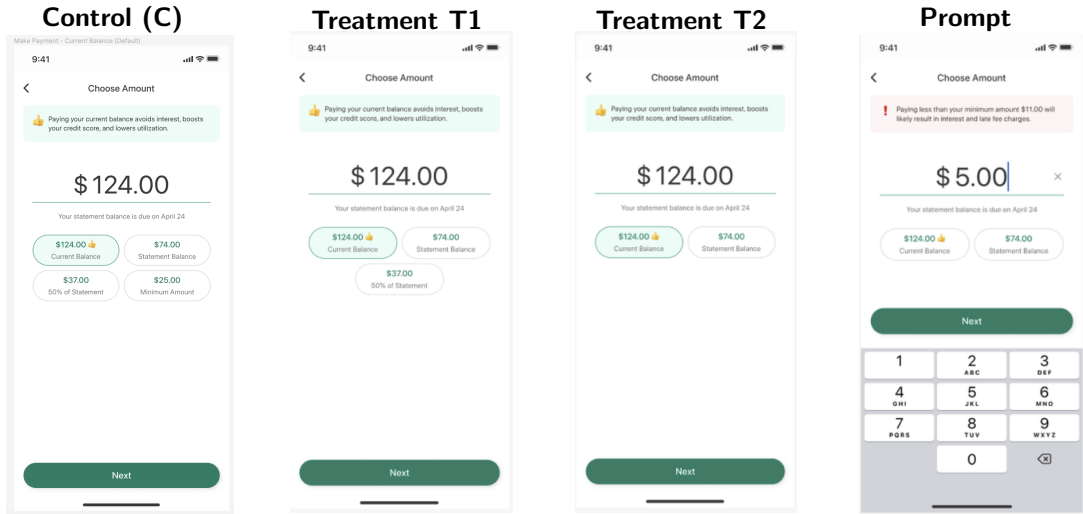
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--------------------------------------	-------------------------------------

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Field Experiment Design



If attempt to pay < minimum, a **prompt** reveals minimum payment

Field Experiment Design ($N = 6,714$)

Control (C)

4,542

Make Payment - Current Balance (Default)

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Prompt

9:41

Choose Amount

! Paying less than your minimum amount \$11.00 will likely result in interest and late fee charges.

\$5.00

Your statement balance is due on April 24

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\$74.00 Statement Balance

Next

1 2 3
4 5 6
7 8 9
0

T1 (vs C): Effect of *shrouding* minimum payment

T2 (vs C): Effect of *shrouding* minimum payment & 50% statement balance options

Field Experiment Design ($N = 6,714$)

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T1 (vs C): Effect of *shrouding* minimum payment

T2 (vs C): Effect of *shrouding* minimum payment & 50% statement balance options

T1 (vs T2): Effect of *adding* 50% option (conditional on shrouding minimum)

Data Provider

U.S. FinTech lender providing credit cards:

- Means: \$290 credit limits, \$200 statement balance, 560 VantageScore.

Below prime (< 660 VantageScore) consumers:

- Group most likely to pay at/near minimum payment, with similar “anchoring” behavior to prime consumers Keys & Wang (2019)
- Group with highest interest rates $\rightarrow$ largest potential gains from $\uparrow$ payments (Drechsler et al., 2025; Consumer Financial Protection Bureau, 2025)

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Datasets

Linked lender's administrative credit card data with (1) Clickstream data on payment choices, (2) Financial transaction data on liquid cash (Plaid), (3) Equifax credit reporting data

Today, focus on regression results from dynamic specification:

$$Y_{i,t} = \sum_{\tau=0}^6 \left[\delta_{\tau}^{T1(vsC)} (D_{\tau} \times T1_i) + \delta_{\tau}^{T2(vsC)} (D_{\tau} \times T2_i) \right] + X_i' \beta + \mu_t + \varepsilon_{i,t} \quad (1)$$

- X_i vector of pre-experiment controls, μ_t statement cycle fixed effects
- Standard errors clustered at consumer-level

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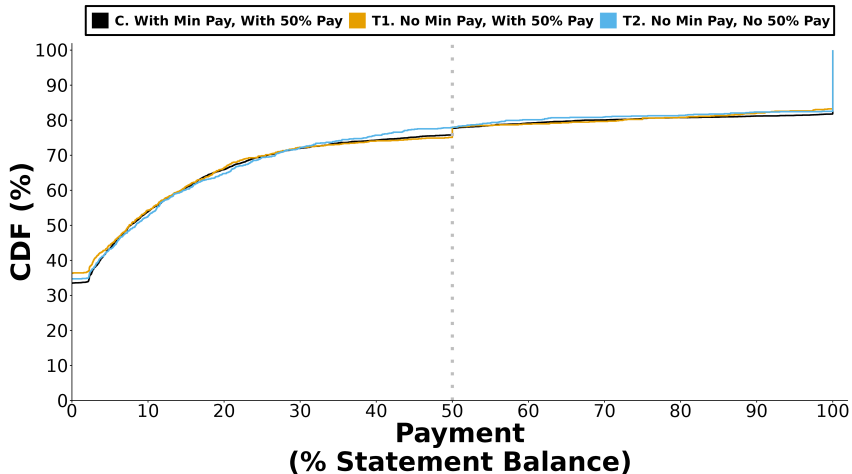
- X_i vector of pre-experiment controls, μ_t statement cycle fixed effects
- Standard errors clustered at consumer-level

Pooled specification to $\uparrow$ power:

$$Y_{i,t} = \delta^{T1(vsC)} T1_i + \delta^{T2(vsC)} T2_i + X_i' \beta + \mu_t + \varepsilon_{i,t} \quad (2)$$

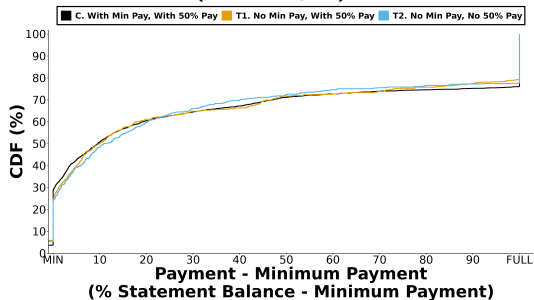
2. Experimental Results

Payment Distribution Shows Little Overall Change After 6 Cycles

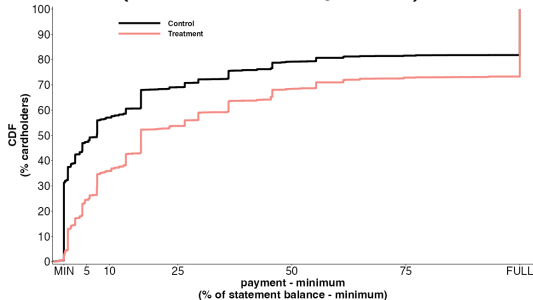


Field Experiment Results In Sharp Contrast To Online Experiments

Field Experiment (This Paper)



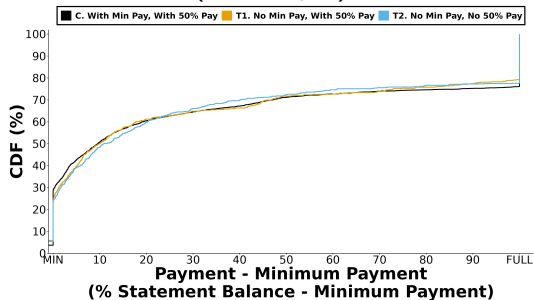
Online Experiment Example (Guttman-Kenney, 2024)



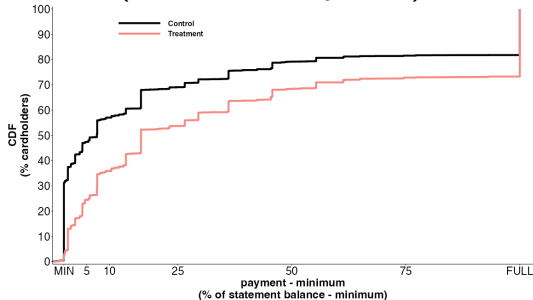
↑ 12.4*** p.p. (s.e. 0.8)
($N = 7,938$, mean 28.1%)

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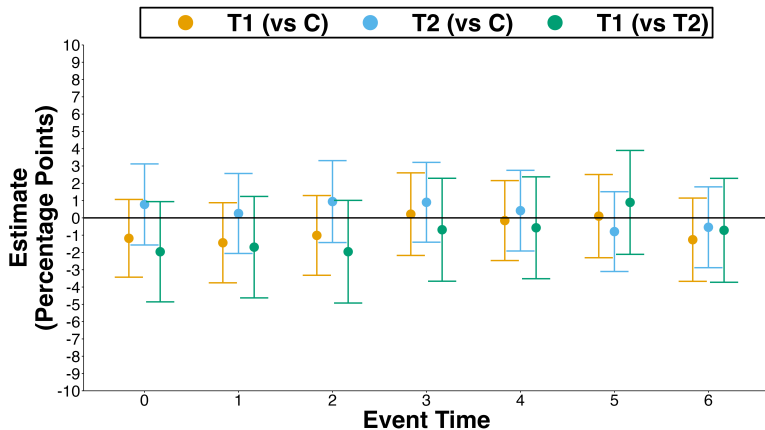
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↑ 12.4*** p.p. (s.e. 0.8)
($N = 7,938$, mean 28.1%)

Challenging for online experiments to recreate the frictions, intertemporal dynamics, & high-stakes incentives that consumers experience managing their finances.

Effects On Primary Outcome: Payments (% Statement Balance)



δ_6^k (percentage points, C mean: 27.8%):

T1 (vs C) -1.26 (s.e. 1.23) **T2 (vs C)** -0.54 (s.e. 1.19) **T1 (vs T2)** -0.72 (s.e. 1.53)

Effects on Payments (% Statement Balance) Pooled Across Cycles 0 to 6

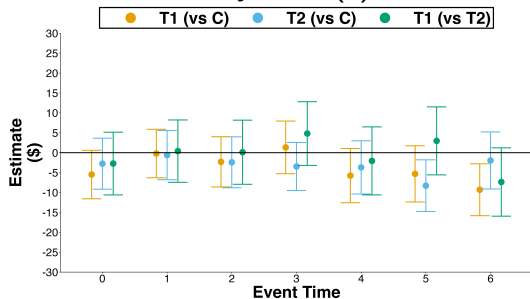
Comparison	Pooled (δ^k)			
	(1)	(2)	(3)	(4)
T1 (vs C)	0.49 (1.06)	0.55 (1.06)	-0.67 (0.81)	-2.39 (1.23)
T2 (vs C)	0.35 (1.05)	0.37 (1.05)	0.29 (0.80)	1.04 (1.16)
T1 (vs T2)	0.14 (1.35)	0.18 (1.35)	-0.96 (1.02)	-3.44* (1.51)
Time F.E.		X	X	X
Consumer Controls			X	
Consumer F.E.				X

(percentage points, C mean: 34.6%)

Other Outcomes & Initial Payment Choices

No Significant $\uparrow$ Payments (\$) Or $\downarrow$ Revolving Debt (\$)

A. Payments (\$)



δ_6^k (percentage points):

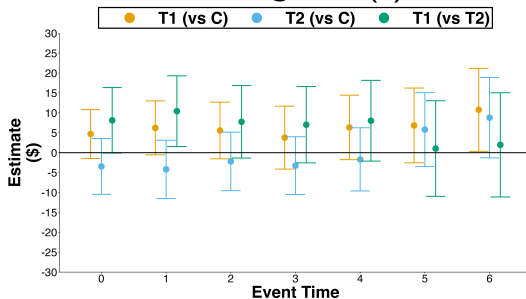
T1 (vs C) $-\$9.41^*$ (s.e. \$3.32)

T2 (vs C) $-\$1.95$ (s.e. \$3.65)

T1 (vs T2) $-\$7.36$ (s.e. \$4.38)

(C mean: \$58.4)

B. Revolving Debt (\$)



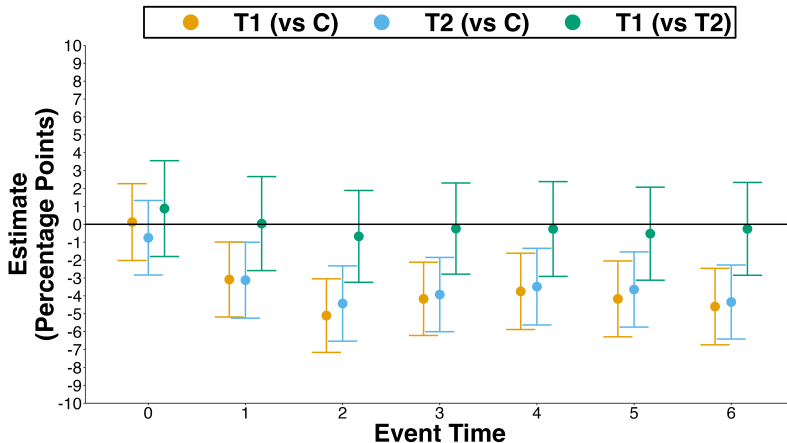
T1 (vs C): $\$10.76^*$ (s.e. \$5.32)

T2 (vs C): $\$8.79$ (s.e. \$5.14)

T1 (vs T2): $\$1.97$ (s.e. \$6.68)

(C mean: \$230.9)

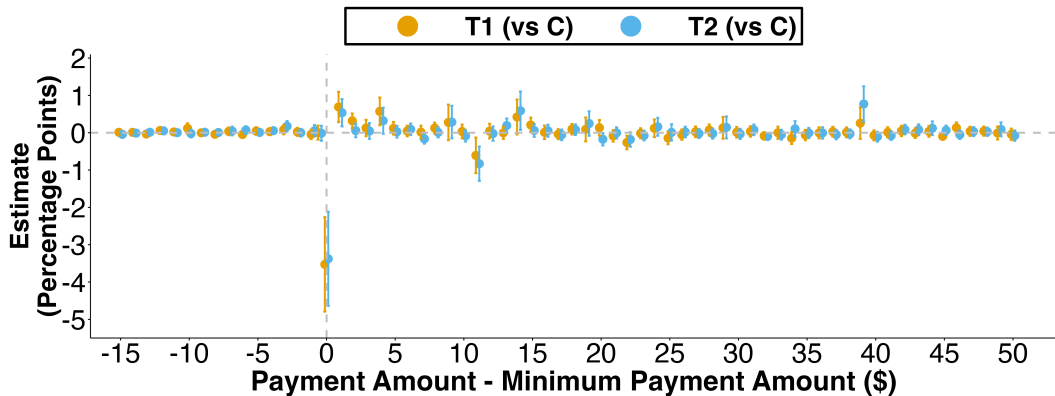
Effects On Payment Distribution: 29% Less Likely To Only Pay The Minimum



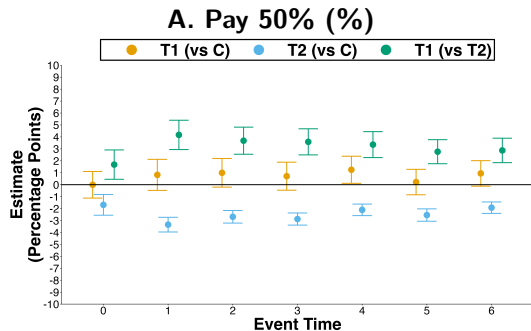
δ_6^k (percentage points, C mean: 15.6%):

T1 (vs C) -4.60*** (s.e. 1.09) **T2 (vs C)** -4.34*** (s.e. 1.06) **T1 (vs T2)** -0.26 (s.e. 1.32)

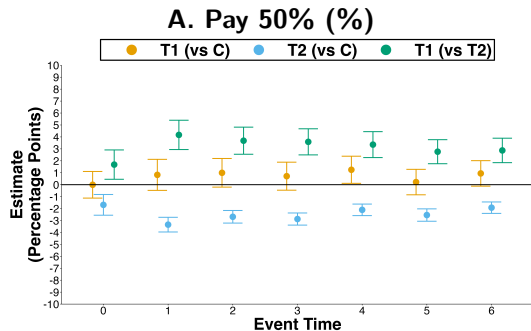
Effects On Payments Proximate To Minimum Payment Amount



50% Payment Option Makes Consumers Initially Less Likely To Pay In Full

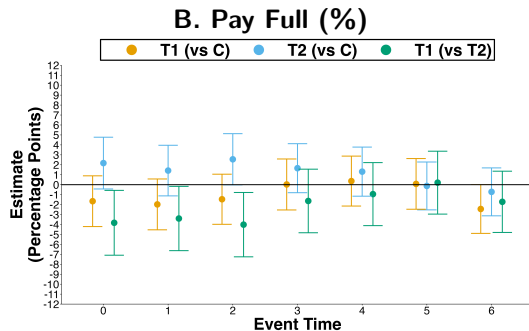


50% Payment Option Makes Consumers Initially Less Likely To Pay In Full



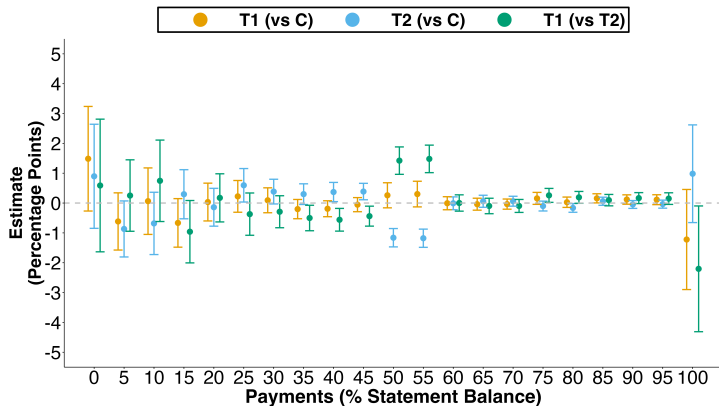
δ_6^k (percentage points):

T1 (vs C) 0.94 (s.e. 0.54)
T2 (vs C) -1.93^{***} (s.e. 0.24)
T1 (vs T2) 2.87^{***} (s.e. 0.52)
(C mean: 1.9%)



T1 (vs C) -2.44 (s.e. 1.25)
T2 (vs C) 0.72 (s.e. 1.23)
T1 (vs T2) -1.72 (s.e. 1.57)
(C mean: 18.2%)

No Evidence Of Shrouding Minimum Payment Causes Payments To Increase

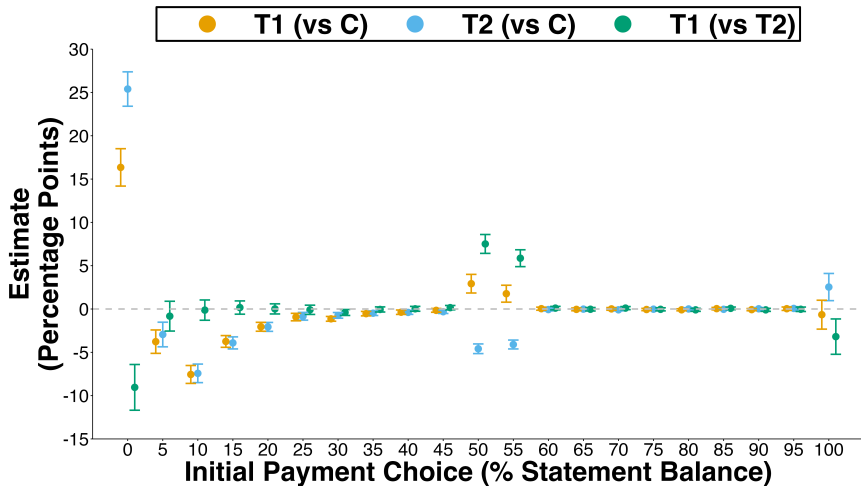


Adding 50% option:

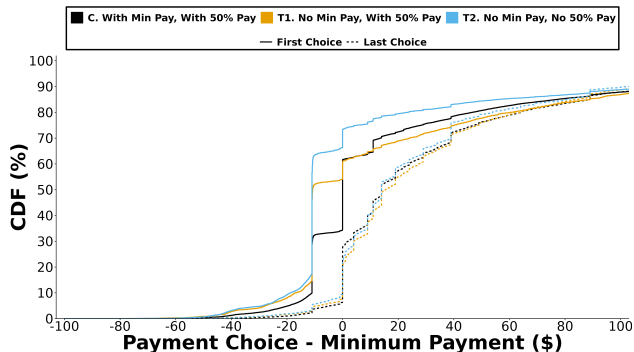
↓ $Pr(\text{Payment} \in \{(30\%, 45\%], 100\%\})$

↑ $Pr(\text{Payment} \in \{(45\%, 55\%]\})$

Clickstream Data: Shrouding Minimum Lowers Initial Payment Choice



CDFs Of Initial Vs. Last Payment Choices Within Payment Cycle



Pr (Initial Payment Choice < Minimum Payment):

T1 (vs C) 19.44*** (s.e. 1.16) T2 (vs C) 30.50*** (s.e. 1.02)

T1 (vs T2) 11.06*** (s.e. 1.38) (percentage points, C mean: 33.7%)

3. Heterogeneity, Distinguishing Anchoring from Targeting

Examine 3 Sources of Heterogeneity (Calculated With Pre-Experiment Data)

1. Liquid Cash
2. Credit Card Utilization
3. (Near Minimum Payers)

Minimum liquid cash balance in 90 days pre-experiment:

- <\$1 for 63% cardholders
- <\$501 for 93% cardholders
- <\$1,001 for 96% cardholders

Minimum liquid cash balance in 90 days pre-experiment:

- <\$1 for 63% cardholders
- <\$501 for 93% cardholders
- <\$1,001 for 96% cardholders

Not sample-specific: Similar for cardholders at large U.K. bank
(Guttman-Kenney et al., 2025)

Heterogeneous Effects By Pre-Trial Liquid Cash Balances

Sample	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Consumers
Liquid Cash <\$1	1.32 (2.28)	1.01 (2.22)	0.31 (2.97)	1,025
Liquid Cash \$1+	-1.36 (0.88)	0.20 (0.86)	-1.56 (1.10)	5,689
Liquid Cash <\$501	-0.46 (1.06)	0.70 (1.06)	-1.16 (1.34)	3,796
Liquid Cash \$501+	-0.80 (1.3)	-0.45 (1.23)	-0.35 (1.60)	2,918
Liquid Cash <\$1,001	-1.11 (0.95)	0.42 (0.95)	-1.53 (1.21)	4,726
Liquid Cash \$1,001+	0.25 (1.58)	-0.22 (1.50)	0.47 (1.96)	1,988
Minimum Liquid Cash <\$1	-1.47 (1.02)	0.68 (1.03)	-2.15 (1.31)	4,230
Minimum Liquid Cash \$1+	-0.25 (1.37)	-0.15 (1.33)	-0.09 (1.70)	2,484
Minimum Liquid Cash <\$501	-0.88 (0.83)	0.61 (0.83)	-1.49 (1.06)	6,252
Minimum Liquid Cash \$501+	1.50 (3.33)	0.43 (3.34)	1.07 (4.07)	462
Minimum Liquid Cash <\$1,001	-0.95 (0.82)	0.21 (0.82)	-1.16 (1.04)	6,422
Minimum Liquid Cash \$1,001+	1.77 (4.46)	5.87 (4.55)	-4.10 (5.77)	292
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

93% cardholders with
minimum liquid cash balance
<\$501 in 90 days pre-experiment

Heterogeneous Effects By Pre-Experiment Credit Card Utilization

Portfolio utilization: 25% (mean) & 36% (median)
16% cardholders with 90%+ portfolio utilization (borrowing constrained)

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 16% cardholders with 90%+ portfolio utilization (borrowing constrained)

Sample	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Consumers
Q1 Card Utilization (<60%)	-1.89 (1.65)	3.70* (1.60)	-5.59** (2.08)	2,238
Q2 Card Utilization (60-91%)	1.57 (1.40)	-1.19 (1.31)	2.76 (1.72)	2,238
Q3 Card Utilization (91%+)	-1.45 (1.15)	-1.30 (1.17)	-0.15 (1.44)	2,238
Q1 Portfolio Utilization (<12%)	-1.68 (1.43)	0.37 (1.47)	-2.05 (1.86)	2,238
Q2 Portfolio Utilization (12-45%)	-2.31 (1.43)	0.94 (1.40)	-3.25 (1.83)	2,238
Q3 Portfolio Utilization (45%+)	1.66 (1.45)	-0.96 (1.36)	2.62 (1.78)	2,238
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

Side Quest: Empirical Support For New Macroeconomic Modeling Approach

- Heterogeneous agent models (“HANK”) include hand-to-mouth consumers with zero liquid wealth **experiencing binding borrowing constraints** (Kaplan & Violante, 2022)
- Lee & Maxted (2025): **binding borrowing constraints inconsistent with SCF data**
 - Consumers often borrow intermediate levels of credit card debt

Side Quest: Empirical Support For New Macroeconomic Modeling Approach

- Heterogeneous agent models (“HANK”) include hand-to-mouth consumers with zero liquid wealth **experiencing binding borrowing constraints** (Kaplan & Violante, 2022)
- Lee & Maxted (2025): **binding borrowing constraints inconsistent with SCF data**
 - Consumers often borrow intermediate levels of credit card debt
- Are Lee & Maxted (2025)’s results due to under-reporting of credit card debt in SCF?(Zinman, 2009) **No!**
 - I also find cardholders with low liquid cash are not borrowing constrained
- My empirical support for Lee and Maxted (2025)’s theory has broad implications,
 - Affects how households respond to fiscal and monetary policy

Distinguishing Anchoring from Targeting

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Anchors: *Irrelevant* numbers influence choices e.g., thinking of your (randomly-assigned) last two digits of SSN influences choices (Tversky & Kahneman, 1974; Ariely et al., 2003)

Targets (reference-dependent preferences): Consumers exert effort to meet/exceed *relevant* numbers e.g., marathon time goals (Heath, Larrick, & Wu, 1999; Allen et al., 2017)

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Distinguish using “**VPA**” **Value Proximity Analysis** (Bartels, Herzog, & Sussman, 2024)

VPA Predicts:

Anchoring: $\alpha \approx 50\%$ (symmetric), $\gamma > 0$ if low anchor, $\gamma < 0$ if high anchor, $\beta^{T1}, \beta^{T2}, \theta^{T1}, \theta^{T2} \neq 0$

Targeting: $\alpha \approx 100\%$ (asymmetric), $\gamma < 0$, $\beta^{T1}, \beta^{T2}, \theta^{T1}, \theta^{T2} = 0$

$$P(Y_i \geq F) = \alpha + \beta^{T1} T1_i + \beta^{T2} T2_i + \gamma Proximity_i + \theta^{T1} (Proximity_i \times T1) + \theta^{T2} (Proximity_i \times T2) + \varepsilon_i$$

VPA Results Inconsistent With Anchoring

	Minimum Payment	50% Statement Balance	Statement Balance
	(1)	(2)	(3)
Intercept	98.510*** (1.411)	76.831*** (2.718)	89.091*** (2.308)
T1	-1.403 (2.121)	5.290 (3.645)	1.380 (3.244)
T2	-2.840 (2.212)	-7.568 (3.986)	3.377 (3.052)
Proximity	-0.406*** (0.028)	-0.584*** (0.042)	-0.798*** (0.037)
Proximity x T1	-0.005 (0.041)	-0.044 (0.057)	-0.012 (0.052)
Proximity x T2	0.008 (0.042)	0.078 (0.062)	-0.016 (0.050)
Observations	300	300	300
Adjusted R ²	0.647	0.773	0.873

Evidence Against Credit Card Anchoring

1. Shrouding minimum payment does not increase payments
2. Initial payment choices are lower when minimum payment is shrouded
3. Value Proximity Analysis results inconsistent with anchoring
4. Heterogeneity analysis shows no increase in payments for near minimum payers

(+ shrouding autopay minimum payment option also ineffective, Guttman-Kenney et al., 2025)

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Instead, makes sense for cardholders to target minimum payment, given:

- Frequently low liquid cash balances
- Discontinuity in economic payoffs generating observed bunching in payments

“An issue that has received relatively little attention in the literature is that kinks and notches may represent reference points” (Kleven, 2016)

Conclusions

Key Findings

- Shrouding the minimum payment does *not* increase in credit card payments
- Adding 50% payment option also does *not* increase credit card payments
- Credit cardholders frequently have low liquid cash balances (& not at borrowing constraint)

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Key Implications

- **Credit Cards:** Below prime credit cardholders frequently have low liquid cash balances.
 - Minimum payment amount is (relevant) target **not** (irrelevant) anchor
- **Behavioral Economics:** Consumers “hack” active choice nudge
- **Macroeconomic Models:** Hand-to-mouth consumers often not at borrowing constraint
- **Survey Methods:** Use online experiments with caution!

Thank you!



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Pre-Experiment Controls

Linear controls:

- Age
- Credit Score
- Card Tenure
- Pre-experiment values of all of the primary and secondary outcomes
- Number of open credit cards held in their portfolio
- Total credit card portfolio balances
- Total credit card portfolio credit limits

Fixed effects:

- Interest rate
- Credit limit
- Autopay type

Primary and Secondary Outcomes

Primary Outcome: 1. Payments (%) = $\sum \text{Payments} / \text{Statement Balance}$)

Secondary Outcomes:

2. Payments (\$)
3. Statement Balance (\$)
4. Spending (\$)
5. Spending (%)
6. Revolving Debt (\$)
7. Credit Limit (\$)
8. Miss Pay (%)
9. Pay Min (%)
10. Pay 50% (%)
11. Pay Full (%)
12. Pay Current Balance (%)

Attrition (3, 6 Cycles)

	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)
3	-0.23 (0.14)	-0.12 (0.18)	-0.12 (0.18)
6	-0.09 (0.64)	0.00 (0.70)	-0.09 (0.83)
Time F.E.	X	X	X
Consumer Controls	X	X	X

Pre-Experiment Summary Statistics

Outcome	C Mean (1)	T1 Mean (2)	T2 Mean (3)	T1 - C (4)	T2 - C (5)
Age	45.51	45.68	45.84	0.17 (0.38)	0.33 (0.39)
Credit Score	561.70	561.06	562.10	[0.652] -0.64 (2.06)	[0.390] 0.40 (2.10)
APR (%)	15.71	15.72	15.72	[0.757] 0 (0.01)	[0.847] 0 (0.01)
Tenure (days)	265.58	257.53	262.24	[0.826] -8.05 (5.10)	[0.85] -3.34 (5.10)
Autopayer (%)	41.19	43.59	39.69	[0.114] 2.39 (1.68)	[0.512] -1.50 (1.65)
Payments (%)	45.06	48.00	44.34	[0.154] 2.95 (1.44)	[0.362] -0.72 (1.39)
Payments	77.07	85.00	76.90	[0.040] 7.93 (4.21)	[0.608] -0.08 (3.79)
Statement Balance	207.94	206.38	208.75	[0.060] -1.55 (7.57)	[0.984] 0.82 (7.41)
Spending	98.13	99.81	101.3	[0.837] 1.68 (4.67)	[0.912] 3.17 (4.82)
Spending (%)	50.11	51.95	50.25	[0.719] 1.84 (1.46)	[0.510] 0.14 (1.44)
Revolving Debt	149.09	144.62	149.98	[0.207] -4.47 (7.15)	[0.925] 0.89 (6.98)
Credit Limit	292.09	287.64	294.11	[0.532] -4.45 (8.16)	[0.898] 2.02 (8.13)
Miss Pay (%)	17.00	14.59	15.88	[0.585] -2.41 (1.21)	[0.803] -1.12 (1.24)
Pay Min (%)	17.48	17.38	16.79	[0.047] -0.10 (1.29)	[0.365] -0.69 (1.26)
Pay 50% (%)	3.74	3.07	4.20	[0.937] -0.68 (0.60)	[0.583] 0.45 (0.67)
Pay Full (%)	31.73	34.39	30.02	[0.257] 2.66 (1.61)	[0.497] -1.71 (1.55)
Pay Current Balance (%)	9.25	11.15	9.67	[0.098] 1.91 (1.05)	[0.270] 0.42 (0.99)
				[0.070] [0.668]	

T-Tests (3 Cycles)

Outcome	C Mean (1)	T1 Mean (2)	T2 Mean (3)	T1 - C (4)	T2 - C (5)	T1 - T2 (6)
Payments (%)	32.92	34.40	33.83	1.48 (1.39) [0.285]	0.92 (1.37) [0.503]	0.57 (1.76) [0.747]
Payments	59.01	63.18	55.75	4.16 (4.00) [0.297]	-3.26 (3.61) [0.367]	7.42 (4.81) [0.123]
Statement Balance	231.01	233.85	225.52	2.84 (7.93) [0.720]	-5.49 (7.65) [0.473]	8.33 (9.89) [0.400]
Spending	71.94	73.64	67.50	1.70 (4.54) [0.708]	-4.43 (4.27) [0.299]	6.14 (5.57) [0.271]
Spending (%)	36.03	38.04	36.55	2.01 (1.43) [0.158]	0.52 (1.42) [0.714]	1.49 (1.82) [0.412]
Revolving Debt	187.21	187.88	182.58	0.67 (7.70) [0.931]	-4.63 (7.48) [0.536]	5.30 (9.65) [0.583]
Credit Limit	320.21	319.02	321.35	-1.20 (8.75) [0.891]	1.14 (8.64) [0.895]	-2.34 (11.02) [0.832]
Miss Pay (%)	28.82	29.58	29.09	0.76 (1.55) [0.622]	0.28 (1.53) [0.857]	0.49 (1.96) [0.803]
Pay Min (%)	16.20	11.63	11.62	-4.57*** (1.12) [0.000]	-4.58*** (1.11) [0.000]	0.01 (1.38) [0.995]
Pay 50% (%)	2.72	3.35	0	0.63 (0.60) [0.293]	-2.72*** (0.24) [0.000]	3.35*** (0.55) [0.000]
Pay Full (%)	22.76	24.19	24.25	1.42 (1.45) [0.326]	1.48 (1.44) [0.303]	-0.06 (1.84) [0.974]
Pay Current Balance (%)	5.90	5.86	6.50	-0.04 (0.80) [0.960]	0.60 (0.82) [0.470]	-0.64 (1.03) [0.539]

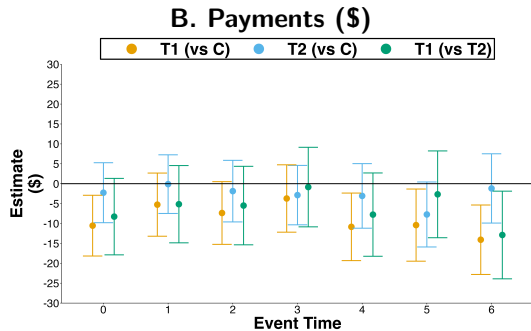
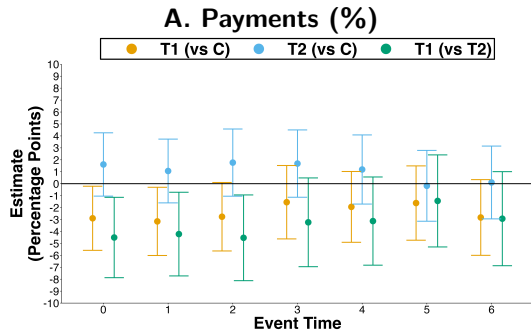
T-Tests (6 Cycles)

Outcome	C Mean (1)	T1 Mean (2)	T2 Mean (3)	T1 - C (4)	T2 - C (5)	T1 - T2 (6)
Payments (%)	27.75	27.52	27.19	-0.23 (1.31)	-0.56 (1.29)	0.33 (1.65)
Payments	58.41	51.53	57.07	[0.864] -6.89 (3.62)	[0.664] -1.34 (3.95)	[0.839] -5.55 (4.73)
Statement Balance	278.41	282.85	283.45	[0.057] 4.44 (8.81)	[0.734] 5.04 (8.52)	[0.241] -0.60 (11.05)
Spending	65.25	62.05	62.15	[0.614] -3.20 (4.59)	[0.554] -3.10 (4.28)	[0.957] -0.10 (5.59)
Spending (%)	27.32	28.59	27.99	[0.486] 1.27 (1.34)	[0.468] 0.66 (1.32)	[0.986] 0.61 (1.70)
Revolving Debt	230.91	239.63	240.26	[0.346] 8.72 (8.59)	[0.616] 9.35 (8.25)	[0.721] -0.63 (10.74)
Credit Limit	366.11	362.14	369.84	[0.310] -3.97 (9.86)	[0.257] 3.73 (9.76)	[0.953] -7.70 (12.47)
Miss Pay (%)	35.97	39.83	37.91	[0.687] 3.86 (1.69)	[0.702] 1.94 (1.67)	[0.537] 1.92 (2.14)
Pay Min (%)	15.62	10.56	10.46	[0.023] -5.06*** (1.11)	[0.246] -5.16*** (1.10)	[0.371] 0.10 (1.35)
Pay 50% (%)	1.85	2.71	0.10	[0.000] 0.86 (0.55)	[0.000] -1.76*** (0.23)	[0.940] 2.62*** (0.52)
Pay Full (%)	18.24	16.96	17.37	[0.116] -1.28 (1.31)	[0.000] -0.87 (1.31)	[0.000] -0.41 (1.66)
Pay Current Balance (%)	4.06	3.78	3.74	[0.328] -0.28 (0.67)	[0.509] -0.31 (0.66)	[0.803] 0.04 (0.84)
				[0.678]	[0.636]	[0.965]

Dynamic Effects on Payments (% Statement Balance) After 3 & 6 Cycles

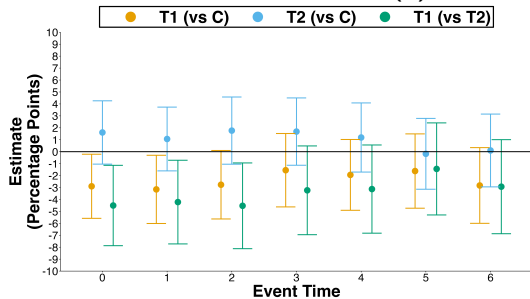
Comparison	Dynamic (δ_3^k)					Dynamic (δ_6^k)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T1 (vs C)	-0.21 (1.33)	1.47 (1.39)	0.22 (1.22)	-1.55 (1.57)	-7.09*** (1.26)	0.01 (1.30)	-1.26 (1.23)	-2.83 (1.62)
T2 (vs C)	-0.78 (1.31)	1.00 (1.37)	0.90 (1.18)	1.68 (1.44)	-7.42*** (1.24)	-0.38 (1.29)	-0.54 (1.19)	0.10 (1.56)
T1 (vs T2)	0.57 (1.75)	0.47 (1.75)	-0.69 (1.52)	-3.23 (1.90)	0.33 (1.64)	0.40 (1.64)	-0.72 (1.53)	-2.93 (2.01)
Time F.E.		X	X	X		X	X	X
Consumer Controls			X				X	
Consumer F.E.				X				X

With Consumer F.E.: Payments

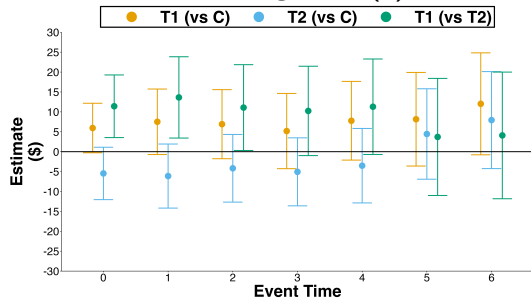


With Consumer F.E.: Statement Balance (\$) & Revolving Debt (\$) 🚀

A. Statement Balance (\$)



B. Revolving Debt (\$)



Effects on Secondary Outcomes After 3 Cycles

Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)
Payments	1.33 (3.37)	-3.48 (3.08)	4.81 (4.08)
Statement Balance	5.10 (3.65)	-4.48 (3.38)	9.57* (4.48)
Spending	-2.69 (3.68)	-4.57 (3.60)	1.88 (4.59)
Spending (%)	0.09 (1.20)	0.39 (1.19)	-0.31 (1.51)
Revolving Debt	3.79 (4.03)	-3.25 (3.70)	7.03 (4.90)
Credit Limit	0.39 (2.20)	-0.50 (2.05)	0.89 (2.75)
Miss Pay (%)	1.42 (1.47)	0.45 (1.44)	0.98 (1.85)
Pay Min (%)	-4.17*** (1.04)	-3.93*** (1.06)	-0.24 (1.3)
Pay 50% (%)	0.71 (0.60)	-2.88*** (0.26)	3.59*** (0.56)
Pay Full (%)	0.02 (1.31)	1.66 (1.26)	-1.64 (1.63)
Pay Current Balance (%)	-0.32 (0.78)	0.60 (0.80)	-0.92 (1.01)
Time F.E.	X	X	X
Consumer Controls	X	X	X

Robustness Effects on Secondary Outcomes After 3 Cycles

Outcome	T1 (vs C)			T2 (vs C)			T1 (vs T2)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Payments	-0.07 (3.84)	4.03 (4.00)	-3.72 (4.32)	-7.50* (3.44)	-3.09 (3.61)	-2.87 (3.81)	7.42 (4.81)	7.12 (4.82)	-0.85 (5.10)
Statement Balance	-3.34 (7.84)	2.45 (7.89)	4.61 (4.45)	-11.67 (7.55)	-6.00 (7.63)	-5.67 (3.79)	8.33 (9.89)	8.45 (9.87)	10.28* (5.19)
Spending	1.07 (4.35)	1.07 (4.53)	-0.42 (4.80)	-5.07 (4.06)	-4.77 (4.27)	-8.11 (4.92)	6.14 (5.57)	5.84 (5.58)	7.69 (6.14)
Spending (%)	2.46 (1.37)	1.93 (1.43)	-0.01 (1.60)	0.97 (1.36)	0.42 (1.42)	0.26 (1.58)	1.49 (1.82)	1.51 (1.81)	-0.27 (2.02)
Revolving Debt	-1.39 (7.58)	0.23 (7.65)	5.18 (4.82)	-6.69 (7.36)	-5.17 (7.46)	-5.06 (4.36)	5.30 (9.65)	5.39 (9.62)	10.24 (5.73)
Credit Limit	-5.61 (8.72)	-2.84 (8.68)	2.18 (2.68)	-3.28 (8.61)	-0.26 (8.56)	-0.84 (2.45)	-2.34 (11.02)	-2.58 (10.94)	3.02 (3.31)
Miss Pay (%)	2.79 (1.46)	0.84 (1.53)	3.24 (1.78)	2.30 (1.44)	0.38 (1.51)	1.29 (1.69)	0.49 (1.96)	0.46 (1.94)	1.95 (2.20)
Pay Min (%)	-4.76*** (1.05)	-4.70*** (1.12)	-4.62*** (1.42)	-4.77*** (1.04)	-4.69*** (1.11)	-3.84** (1.46)	0.01 (1.38)	-0.01 (1.37)	-0.78 (1.80)
Pay 50% (%)	0.75 (0.56)	0.68 (0.60)	1.36 (0.8)	-2.59*** (0.12)	-2.74*** (0.25)	-3.23*** (0.71)	3.35*** (0.55)	3.43*** (0.56)	4.59*** (0.95)
Pay Full (%)	0.36 (1.39)	1.33 (1.45)	-1.45 (1.77)	0.42 (1.38)	1.57 (1.44)	3.22* (1.60)	-0.06 (1.84)	-0.24 (1.84)	-4.66* (2.13)
Pay Current Balance (%)	-0.65 (0.74)	-0.08 (0.80)	-1.83 (1.23)	-0.01 (0.77)	0.55 (0.83)	0.18 (1.19)	-0.64 (1.03)	-0.63 (1.04)	-2.01 (1.55)
Time F.E.		X	X		X	X		X	X
Consumer F.E.			X			X			X

Effects on Secondary Outcomes After 6 Cycles

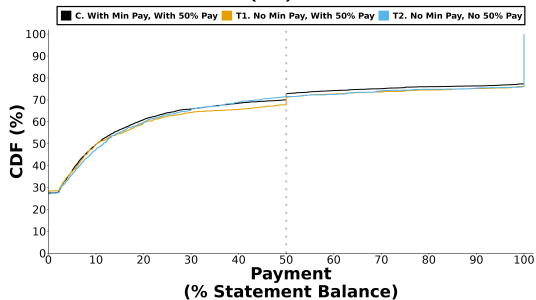
Outcome	T1 (vs C)		T2 (vs C)		T1 (vs T2)	
	(1)	(2)	(3)	(4)	(5)	(6)
Payments	-9.31** (3.32)	-14.08*** (4.45)	-1.95 (3.65)	-1.19 (4.44)	-7.36 (4.38)	-12.89* (5.62)
Statement Balance	6.03 (4.96)	5.54 (6.04)	3.92 (4.94)	3.77 (5.69)	2.11 (6.33)	1.78 (7.46)
Spending	-6.83 (4.27)	-4.28 (5.48)	-3.27 (3.99)	-6.73 (5.51)	-3.56 (5.27)	2.45 (7.00)
Spending (%)	-0.41 (1.28)	-0.26 (1.75)	0.71 (1.22)	0.52 (1.64)	-1.12 (1.6)	-0.78 (2.16)
Revolving Debt	10.76* (5.32)	12.02 (6.53)	8.79 (5.14)	7.94 (6.22)	1.97 (6.68)	4.09 (8.12)
Credit Limit	-2.15 (4.33)	-0.14 (4.97)	0.35 (4.17)	0.08 (4.74)	-2.50 (5.47)	-0.22 (6.22)
Miss Pay (%)	4.39** (1.63)	5.70*** (1.91)	1.92 (1.61)	2.71 (1.88)	2.47 (2.06)	2.99 (2.41)
Pay Min (%)	-4.60*** (1.09)	-4.82*** (1.56)	-4.34*** (1.06)	-3.99** (1.50)	-0.26 (1.32)	-0.84 (1.93)
Pay 50% (%)	0.94 (0.54)	1.60* (0.80)	-1.93*** (0.24)	-2.20*** (0.71)	2.87*** (0.52)	3.80*** (0.95)
Pay Full (%)	-2.44 (1.25)	-3.80* (1.78)	-0.72 (1.23)	0.66 (1.70)	-1.72 (1.57)	-4.46* (2.21)
Pay Current Balance (%)	-0.48 (0.66)	-1.96 (1.17)	-0.33 (0.66)	-0.79 (1.17)	-0.16 (0.84)	-1.18 (1.49)
Time F.E.	X	X	X	X	X	X
Consumer Controls	X		X		X	
Consumer F.E.		X		X		X

Pooled Effects on Secondary Outcomes

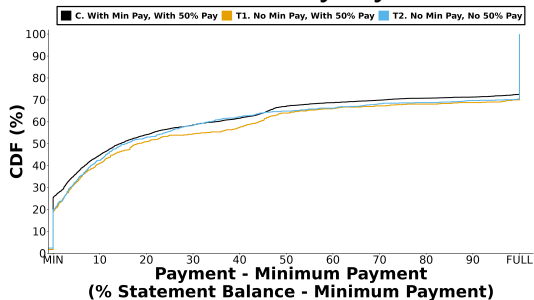
Outcome	T1 (vs C)		T2 (vs C)		T1 (vs T2)	
	(1)	(2)	(3)	(4)	(5)	(6)
Payments	-3.83 (2.19)	-8.85* (3.43)	-3.31 (2.13)	-2.73 (3.12)	-0.52 (2.74)	-6.12 (4.15)
Statement Balance	4.39 (2.42)	3.85 (3.63)	-2.29 (2.35)	-3.40 (3.08)	6.68 (3.04)	7.25 (4.21)
Spending	-2.49 (2.13)	-0.25 (3.93)	-2.32 (2.10)	-5.87 (4.11)	-0.17 (2.70)	5.61 (5.12)
Spending (%)	-0.38 (0.71)	-0.46 (1.29)	0.72 (0.71)	0.56 (1.25)	-1.10 (0.9)	-1.02 (1.61)
Revolving Debt	6.29* (2.78)	7.61 (3.95)	-0.10 (2.72)	-1.81 (3.64)	6.39 (3.50)	9.42* (4.77)
Credit Limit	-0.89 (1.72)	0.91 (2.26)	-0.09 (1.64)	-0.41 (2.11)	-0.80 (2.17)	1.32 (2.80)
Miss Pay (%)	1.72 (0.90)	3.43* (1.34)	1.29 (0.90)	2.18 (1.30)	0.43 (1.15)	1.26 (1.66)
Pay Min (%)	-3.53*** (0.65)	-3.94*** (1.17)	-3.38*** (0.64)	-3.29** (1.18)	-0.15 (0.80)	-0.65 (1.49)
Pay 50% (%)	0.70* (0.29)	1.35* (0.62)	-2.46*** (0.15)	-2.80*** (0.67)	3.16*** (0.28)	4.15*** (0.81)
Pay Full (%)	-1.00 (0.85)	-2.43 (1.42)	1.20 (0.82)	2.75* (1.30)	-2.20* (1.07)	-5.18*** (1.72)
Pay Current Balance (%)	-0.67 (0.40)	-2.17* (1.01)	0.26 (0.42)	-0.16 (0.97)	-0.93 (0.52)	-2.01 (1.26)
Time F.E.	X	X	X	X	X	X
Consumer Controls	X		X		X	
Consumer F.E.		X		X		X

Distribution of Payments, After Three Cycles 🚀

A. CDF Payments (%)

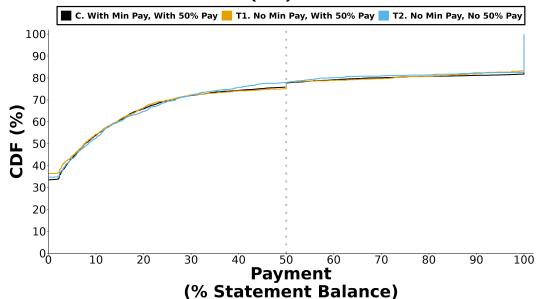


B. CDF Excess Payments (\$), Conditional On Any Payment

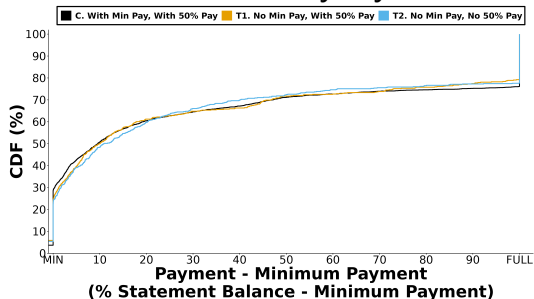


Distribution of Payments, After Six Cycles

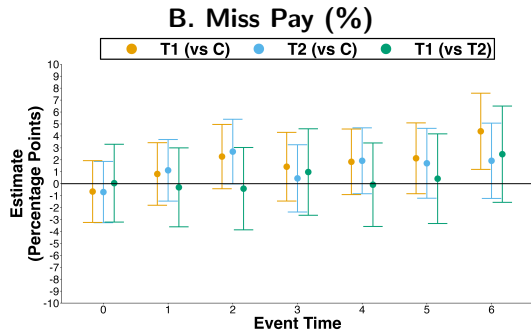
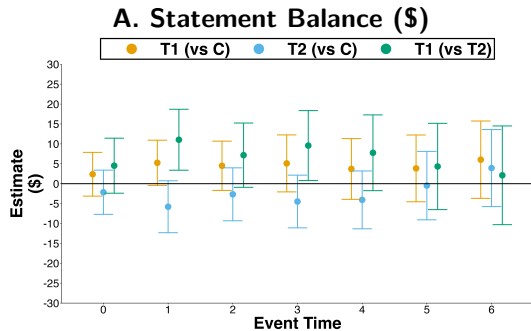
A. CDF Payments (%)



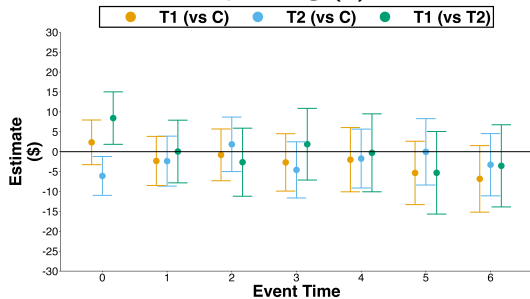
B. CDF Excess Payments (\$), Conditional On Any Payment



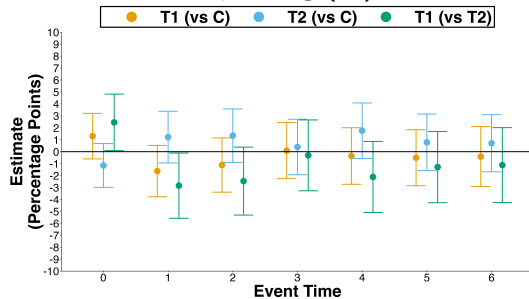
Statement Balance (\$), Miss Pay (%) 🚀



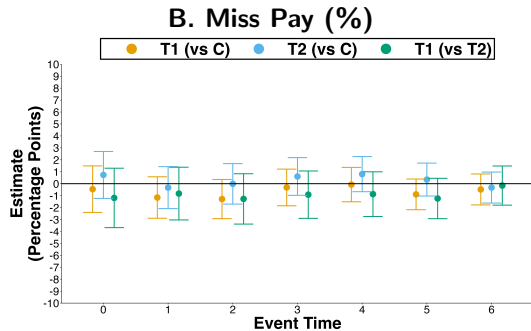
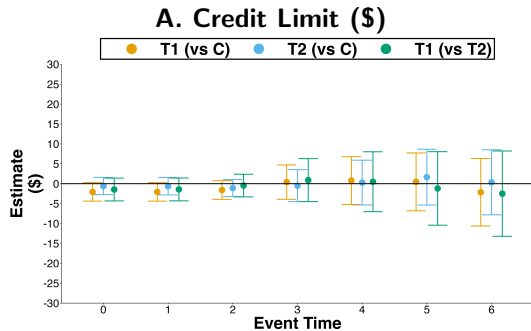
A. Spending (\$)



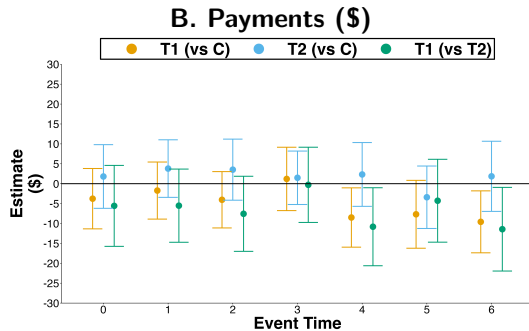
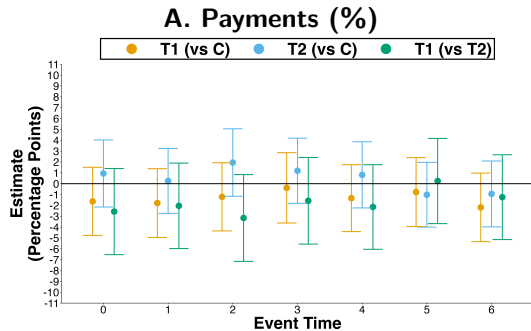
B. Spending (%)



Credit Limit (\$) and Pay Current Balance (%) 🚀



No Autopay Subsample: Payments 🚀



No Autopay Subsample: Dynamic Effects After 6 Cycles 🚀

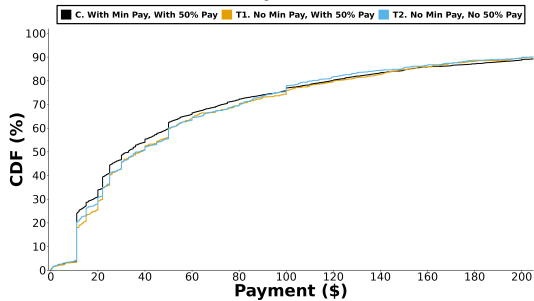
Outcome	T1 (vs C)		T2 (vs C)		T1 (vs T2)	
	(1)	(2)	(3)	(4)	(5)	(6)
Payments (%)	-2.18 (1.61)	-2.97 (2.08)	-0.94 (1.55)	-0.24 (2.02)	-1.24 (1.99)	-2.73 (2.58)
Payments	-9.58* (3.97)	-10.48* (5.08)	1.85 (4.50)	3.84 (5.37)	-11.44* (5.36)	-14.31* (6.51)
Statement Balance	5.66 (5.8)	3.30 (6.93)	8.07 (5.75)	8.87 (6.78)	-2.41 (7.42)	-5.57 (8.82)
Spending	-9.12 (5.10)	-7.76 (6.94)	1.15 (5.00)	0.10 (6.69)	-10.27 (6.48)	-7.86 (8.65)
Spending (%)	-1.12 (1.73)	-0.42 (2.37)	0.03 (1.58)	-0.39 (2.15)	-1.15 (2.11)	-0.03 (2.89)
Revolving Debt	10.31 (6.09)	9.85 (7.46)	9.13 (6.04)	7.11 (7.43)	1.18 (7.76)	2.74 (9.47)
Credit Limit	-3.17 (5.2)	-2.24 (5.78)	5.93 (5.20)	6.46 (5.89)	-9.09 (6.73)	-8.70 (7.50)
Miss Pay (%)	5.03* (2.21)	5.38* (2.55)	2.01 (2.12)	2.77 (2.42)	3.02 (2.75)	2.61 (3.16)
Pay Min (%)	-4.99*** (1.24)	-5.17*** (1.93)	-5.63*** (1.18)	-6.05*** (1.77)	0.64 (1.46)	0.88 (2.33)
Pay 50% (%)	0.81 (0.81)	2.22 (1.15)	-2.66*** (0.38)	-3.05*** (1.08)	3.47*** (0.78)	5.26*** (1.41)
Pay Full (%)	-3.84* (1.62)	-3.81 (2.35)	-1.33 (1.58)	0.43 (2.22)	-2.51 (2.01)	-4.24 (2.89)
Pay Current Balance (%)	-0.29 (0.92)	-0.96 (1.53)	-0.45 (0.88)	-0.42 (1.52)	0.16 (1.13)	-0.54 (1.94)
Time F.E.	X	X	X	X	X	X
Consumer Controls	X		X		X	
Consumer F.E.		X		X		X

Effects On Portfolio of Credit Card Statement Balances 🍷

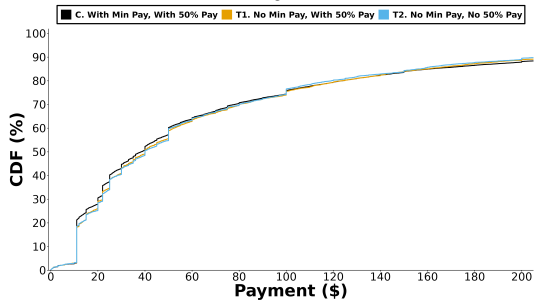
Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean
Portfolio Statement Balances	36.55 (51.85)	6.60 (53.41)	29.94 (66.98)	\$3,681
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

Distribution of Payments (\$) 🚀

A. Cycle 2

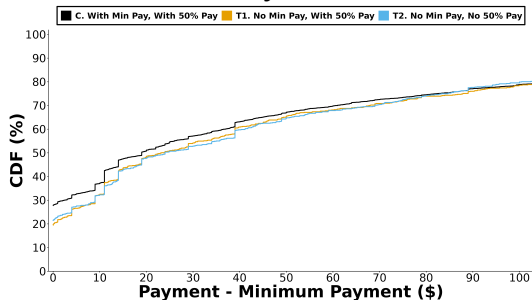


B. Pooled Cycles 0 to 6

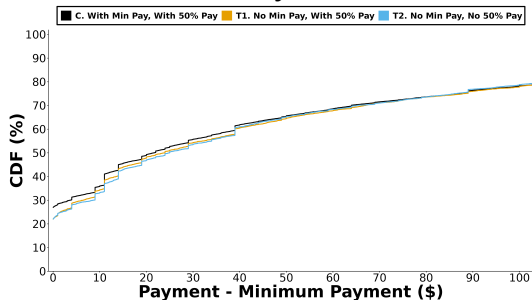


Distribution of Excess Payments (\$)

A. Cycle 2

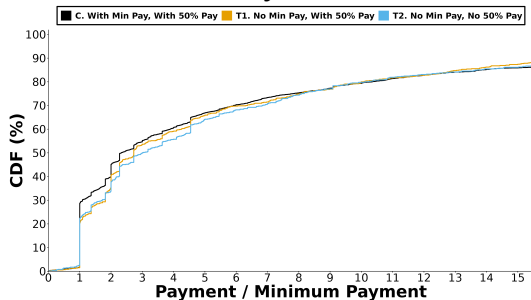


B. Pooled Cycles 0 to 6

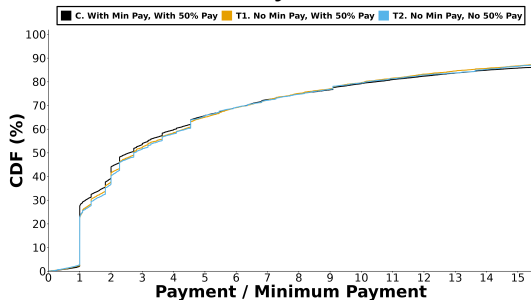


Distribution of Ratio of Payments To Minimum Payment

A. Cycle 2



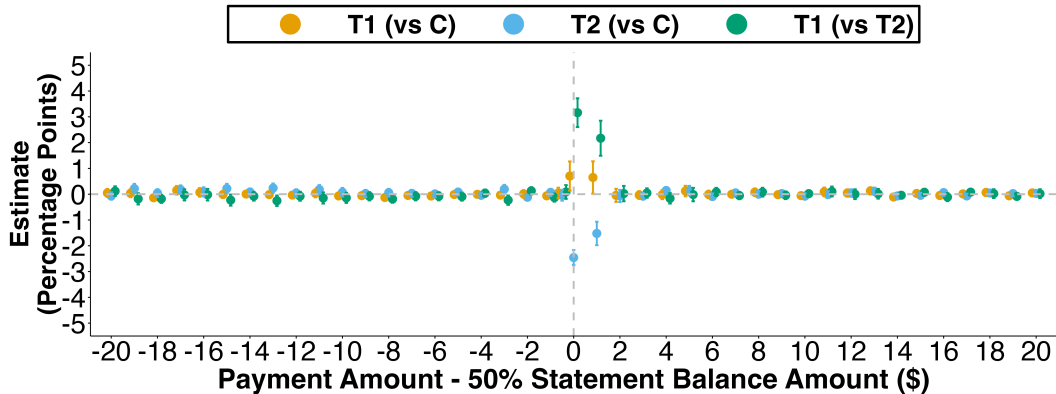
B. Pooled Cycles 0 to 6



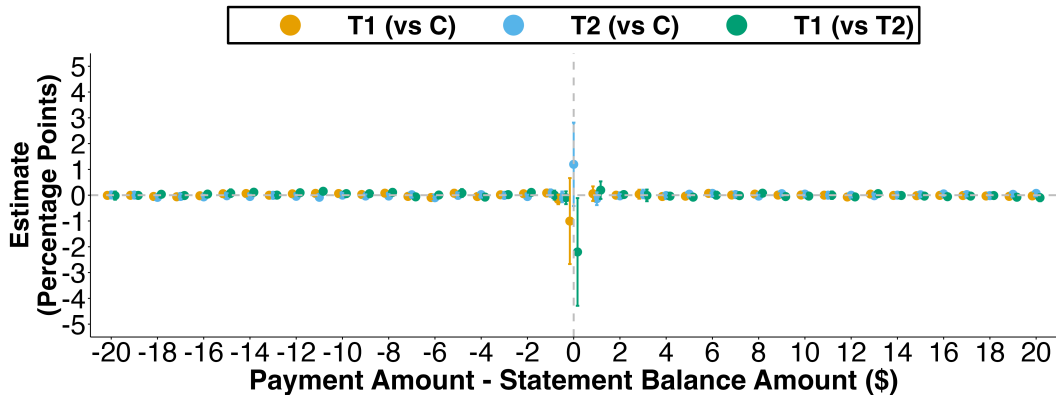
Effects On Heuristic Payment Amounts

Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)
Pay Multiples Of Minimum	-0.76*** (0.25)	-0.94*** (0.23)	0.18 (0.29)
Pay Round Number	2.51*** (0.62)	3.61*** (0.62)	-1.10 (0.80)
Time F.E.	X	X	X
Consumer Controls	X	X	X

Effects On Payments Proximate To 50% Statement Balance Amount 🚀

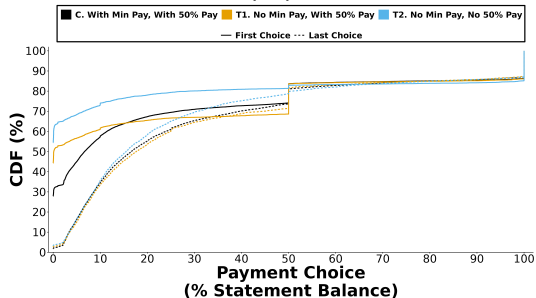


Effects On Payments Proximate To Statement Balance Amount 🚀

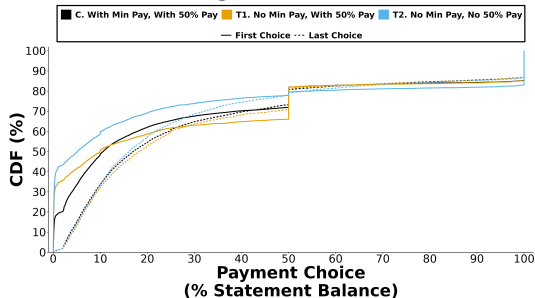


Initial Vs. Last Payment Choices Within Payment Cycle 🚀

A. CDF Initial Payment Choices (%)

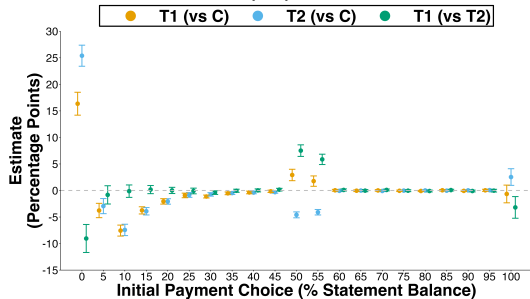


B. CDF Initial Payment Choices (%), Excluding Blanks & Zeros

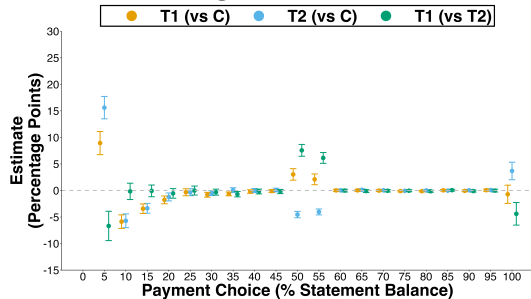


Effects On Initial Payment Choice

A. Initial Payment Choices (%)



B. Initial Payment Choices (%), Excluding Blanks & Zeros



A. Initial Payment Choices

Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)
Initial Payment Choice (% Statement Balance)	-0.90 (0.87)	-4.33*** (0.81)	3.43*** (1.07)
Initial Payment Choice Less Than Minimum Payment	19.44*** (1.16)	30.50*** (1.02)	-11.06*** (1.38)
Number of Payment Choices	0.40*** (0.11)	0.85*** (0.11)	-0.46*** (0.14)
Time F.E.	X	X	X
Consumer Controls	X	X	X

B. Initial Payment Choices, Excluding Blanks & Zeros

Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)
Initial Payment Choice (% Statement Balance)	-0.22 (0.86)	-1.82* (0.83)	1.60 (1.07)
Initial Payment Choice Less Than Minimum Payment	15.59*** (1.08)	23.57*** (1.04)	-7.98*** (1.38)
Number of Payment Choices	0.40*** (0.11)	0.87*** (0.11)	-0.47*** (0.14)
Time F.E.	X	X	X
Consumer Controls	X	X	X

Follow Keys & Wang (2019) Defining Near Minimum Payers 🚀

Anchoring predicts largest results for unconstrained cardholders “near minimum payers”:

$$payments \in \{(Minimum, Minimum + \$50] \setminus Full\}$$

18% my data, 20% in Keys & Wang (2019)

Follow Keys & Wang (2019) Defining Near Minimum Payers 📌

Anchoring predicts largest results for unconstrained cardholders “near minimum payers”:

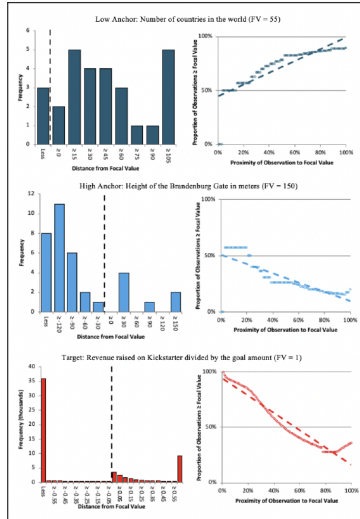
$$\text{payments} \in \{(Minimum, Minimum + \$50] \setminus Full\}$$

18% my data, 20% in Keys & Wang (2019)

Sample	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Consumers
Near Minimum	2.89 (1.73)	-0.56 (1.54)	3.45 (2.07)	1,226
Not Near Minimum	-1.44 (0.92)	0.28 (0.92)	-1.72 (1.17)	5,488
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

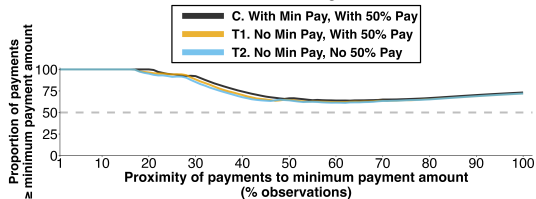
Example Histogram & VPA Charts in Bartels, Herzog, & Sussman (2024) 📍

Example Histograms and VPA Charts

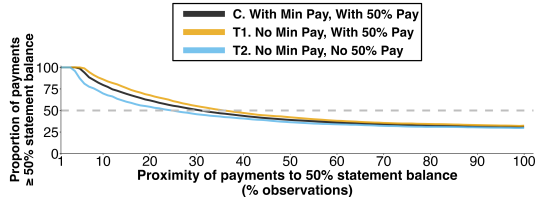


Value Proximity Analysis (VPA): Payments Proximate To Focal Amounts

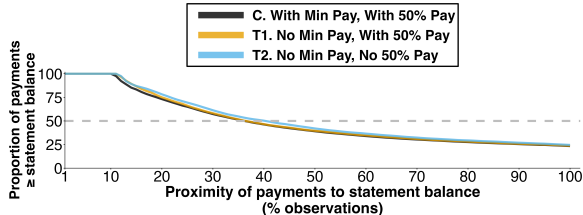
A. Minimum Payment



B. 50% Statement Balance



C. Statement Balance



Reconciling Results With Credit Card Literature 🍷

- Credit card behavior is difficult to shift.
- Understand why suggested payment amounts (e.g., 50%, CARD Act scenarios) ineffective.
 - Intermediate payment values struggle to compete with cardholders targeting paying at least the minimum and the full amount, where strong economic incentives exist.
- Understand why lenders set low minimum payments and why increases to minimum payments can reduce debt, in a way inconsistent with liquidity constraints.

(Beshears et al., 2018; Agarwal et al., 2015; Keys & Wang, 2016, 2019; Seira et al., 2017; Medina, 2021; Kuchler & Pagel, 2021; Adams et al., 2022; Medina & Negrin, 2022; Katz et al., 2024; Castellanos et al., 2025; Guttman-Kenney et al., 2025; Batista et al., 2025; Han & Yin, 2025; Allen et al., 2026)